

A Project Report

On

**DESIGN AND IMPLEMENTATION OF A SOLAR-POWERED
COMMUNITY CHARGING STATION FOR MOBILE PHONES IN
RURAL WOMEN'S SELF-HELP GROUPS.**

Submitted by

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Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRICAL AND ELECTRONICS ENGINEERING

Under the Esteemed Guidance of

Mr. P VICTOR BABU

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DEPARTMENT OF EEE



Department of Electrical and Electronics Engineering

**RAMACHANDRA COLLEGE OF ENGINEERING
(AUTONOMOUS)**

(Approved by AICTE, Permanently Affiliated to JNTUK, Kakinada)

Accredited by NBA and NAAC (A+)

NH-16 Bypass, Vatluru (V), ELURU-534007, Eluru Dist., A.P.

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BONAFIDE CERTIFICATE

This is to certify that this project report entitled “**Design and Implementation of a Solar-Powered Community Charging Station for Mobile Phones in Rural Women's Self-Help Groups.**” is being submitted by **MADU AJAY (22ME1A0234)** in partial fulfillment of **BACHELOR OF TECHNOLOGY** in **ELECTRICAL AND ELECTRONICS ENGINEERING.**

This report is a bonafide work carried out under my guidance and supervision during the academic year 2025-2026 and it has been found worthy of acceptance as per the requirements of the university.

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I am, **MADU AJAY (22ME1A0234)** hereby declare that the project report titled “**Design and Implementation of a Solar-Powered Community Charging Station for Mobile Phones in Rural Women's Self-Help Groups**” has been completed under the supervision of **Mr. P. VICTOR BABU, Assistant Professor, Department of Electrical and Electronics Engineering**. This report is submitted in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ELECTRICAL AND ELECTRONICS ENGINEERING**.

This is a record of work carried out by us, and the results presented in this project have not been reproduced or copied from any source. Furthermore, the results embodied in this project report have not been submitted to any other University or Institute for the award of any other degree or diploma.

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I take this opportunity to express our **sincere gratitude** to all those who have contributed to the successful completion of our research work. Their guidance, support, and encouragement have been invaluable throughout this journey.

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ABSTRACT

This project presents the design, development, and field deployment of a solar-powered Community Charger unit specifically engineered for Self Help Groups (SHGs) operating in rural and semi-urban areas of India. The system addresses a critical gap in energy access by providing a reliable, sustainable, and cost-effective multi-device USB charging infrastructure that operates entirely on solar energy harvested by a 15-watt monocrystalline silicon photovoltaic panel.

The community charger integrates five principal subsystems: a 15W monocrystalline solar panel as the primary energy harvesting source; a 10A Pulse Width Modulation (PWM) solar charge controller for intelligent battery management and protection; a 12V 7.5Ah sealed lead-acid (SLA) battery for energy storage and backup power; two DC-DC buck converter modules (LM2596-based) for regulated 5V voltage conversion; and four USB power bank-style output modules providing the user charging interface.

The assembled system delivers stable 5V USB power to four devices simultaneously, with a measured battery backup autonomy of 2.7 hours under full four-port load. Total component cost is approximately Rs. 2,700, offering a simple payback period of under two months under a nominal user-fee model. The project is designed as a fully replicable, open-documentation community technology solution aligned with India's digital inclusion, rural electrification, and women's empowerment policy objectives.

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INRODUCTION

1.1 General

In the modern world, electricity plays a vital role in supporting daily life, economic development, and technological growth. The increasing use of electronic devices such as mobile phones, computers, and communication systems has led to a significant rise in energy demand. However, in many rural and remote areas, access to reliable electricity remains a major challenge. Despite government efforts to improve electrification, issues such as frequent power outages, voltage fluctuations, and lack of infrastructure continue to affect rural communities. These problems limit the effective use of electrical devices and hinder overall development.

To address these challenges, renewable energy sources have gained considerable importance in recent years. Among all renewable sources, solar energy is one of the most widely used and promising options due to its abundance, availability, and eco-friendly nature. Solar photovoltaic (PV) systems convert sunlight directly into electrical energy without producing pollution or harmful emissions. These systems are easy to install, require minimal maintenance, and can operate independently of the electrical grid, making them highly suitable for rural applications.

Solar energy is particularly beneficial in countries like India, where sunlight is available for most of the year. On average, India receives about 4.5 to 6.5 peak sun hours per day, making solar power a reliable and sustainable energy source. Small-scale solar systems can be effectively used for applications such as lighting, water pumping, and mobile charging in areas where grid electricity is unreliable or unavailable.

The use of standalone solar systems reduces dependency on fossil fuels and minimizes environmental impact. It also helps in reducing electricity costs and ensures energy independence. Furthermore, solar energy systems can be easily scaled based on the requirements, making them flexible for different applications.

Therefore, adopting solar energy-based solutions is an effective way to overcome energy challenges in rural areas. It not only improves access to electricity but also supports sustainable development and enhances the quality of life for people living in underserved regions.

1.2 Charging Station Overview

A solar-powered community charging station is a standalone system designed to provide reliable mobile charging facilities using solar energy. This system is particularly useful in rural areas where electricity supply is inconsistent or completely unavailable. The main objective of the charging station is to ensure that users can charge their mobile phones and other small electronic devices without depending on grid electricity.

The system consists of several important components, including a solar photovoltaic panel, a charge controller, a battery for energy storage, DC–DC buck converters, and USB output modules. The solar panel captures sunlight and converts it into electrical energy in the form of direct current (DC). This energy is then supplied to the charge controller, which regulates the charging process and protects the battery from overcharging, deep discharge, and reverse current.

The battery plays a crucial role in storing the energy generated during the daytime. It acts as a backup source and supplies power during nighttime or when sunlight is not available. The stored energy from the battery is then converted into a stable 5V output using buck converters, which are efficient DC–DC converters used to step down voltage levels. This regulated output is suitable for charging mobile devices.

USB output modules are used as the user interface, allowing multiple users to connect their devices simultaneously. The system is designed to charge multiple phones at the same time, making it suitable for community use in places such as village centers, Self-Help Group meeting halls, and public areas.

The charging station is compact, portable, and easy to install. It requires minimal maintenance and can operate efficiently under varying environmental conditions. Since it uses renewable energy, it reduces dependency on conventional power sources and contributes to environmental protection. Overall, the system provides a reliable, cost-effective, and sustainable solution for mobile charging in rural communities.

1.3 Problem Statement

The following specific challenges faced by Women's Self-Help Group (SHG) members and rural communities highlight the need for an effective and sustainable mobile charging solution:

Frequent grid power outages remain one of the most critical issues in rural regions across states such as Andhra Pradesh, Telangana, Odisha, and Bihar. In many areas, electricity supply is inconsistent, with outages lasting anywhere between 4 to 12 hours daily. This unreliable power supply prevents regular charging of mobile phones, which have become essential tools for communication, financial transactions, and accessing digital services. As a result, individuals are often left with uncharged devices during important situations.

Another major challenge is the absence of dedicated charging infrastructure in common community locations. Places such as village panchayat offices, anganwadi centres, mahila mandal halls, and SHG meeting points typically do not have proper electrical facilities for charging multiple devices. This lack of infrastructure creates inconvenience for users and limits the effective use of mobile technology within the community.

In off-grid or poorly electrified areas, people often depend on alternative energy sources such as diesel generators or kerosene lamps. These methods are not only expensive to operate but also harmful to the environment and human health. Continuous use of such systems leads to increased fuel costs, air pollution, and exposure to harmful emissions, making them unsuitable for long-term use.

Dead or uncharged smartphone batteries create serious problems in daily life. Individuals may miss important calls, fail to complete digital payment transactions through UPI platforms, and lose access to government services or emergency communication. This negatively affects both personal and economic activities, especially in rural areas where digital services are increasingly important.

Additionally, women SHG members often have to travel long distances to nearby towns, shops, or neighbours' houses to charge their phones. This process consumes valuable time and effort, which could otherwise be used for productive activities such as income generation, household management, or participation in SHG meetings. This not only reduces efficiency but also adds to physical strain and inconvenience.

These challenges clearly indicate the need for a reliable, affordable, and eco-friendly charging solution that can be implemented at the community level. A solar-powered charging station effectively addresses these issues by providing a sustainable and accessible source of energy.

1.4 Objectives of the Project

The primary objective of this project is to design and develop a practical, efficient, and sustainable solar-powered community charging station specifically intended for rural Women's Self-Help Groups (SHGs). The system is designed to be compact, portable, and weather-resistant so that it can be easily installed at SHG meeting points, village centers, or other common community locations. The design ensures durability and reliable operation under varying environmental conditions such as heat, dust, and humidity.

Another important objective is to utilize commercially available, low-cost, and easily replaceable components in the system design. By using off-the-shelf components such as solar panels, PWM charge controllers, batteries, buck converters, and USB modules, the system becomes easy to assemble, maintain, and repair without requiring specialized tools or expertise. This approach also ensures that spare parts are readily available in local markets.

Cost-effectiveness is a key consideration in this project. The system is designed to achieve a total cost of under Rs. 3,000, making it affordable for SHGs to procure and implement without depending on external financial support or subsidies. This enhances the feasibility of large-scale adoption in rural areas.

The project also aims to provide simultaneous charging capability for multiple users. The system is designed to support at least four USB output ports, each capable of delivering standard 5V output with current ratings of 1A to 2.1A. This ensures that multiple mobile devices can be charged efficiently at the same time, meeting the needs of community users.

Ensuring reliable operation during non-sunny periods is another important objective. The system includes battery storage to provide backup power, with a target autonomy of at least two hours under full load conditions. This guarantees uninterrupted charging availability even during cloudy weather or after sunset.

Finally, the project aims to create a fully documented and replicable model that can be easily adopted by other SHGs, non-governmental organizations (NGOs), educational institutions, and government agencies. Proper documentation of design, components, assembly, and operation ensures that the system can be reproduced and scaled across different regions, promoting sustainable development and energy access.

1.5 Scope of the Project

The scope of this project covers the complete engineering lifecycle involved in the design and implementation of a solar-powered community charging station for mobile phones. It begins with the conceptual design phase, where the system requirements are identified based on the needs of rural communities and Women's Self-Help Groups (SHGs). This includes defining system specifications such as input power, output voltage, number of charging ports, and backup requirements.

The project includes the development of the electrical schematic design, which involves selecting appropriate components such as the solar panel, charge controller, battery, DC–DC buck converters, and USB output modules. Careful consideration is given to ensure compatibility between components and to achieve efficient energy conversion and storage. Component selection and procurement are carried out using commercially available and cost-effective materials to ensure affordability and ease of maintenance.

The scope also includes mechanical design and assembly of the system. This involves designing a compact, portable, and weather-resistant enclosure to house all components safely. Proper wiring techniques, cable management, and protection mechanisms such as fuses are incorporated to ensure safe and reliable operation. The system is assembled following standard procedures, ensuring correct polarity and secure connections.

Commissioning of the system is another important aspect, which includes verifying all electrical connections, testing individual components, and ensuring that the system operates as intended. Performance testing is conducted under different conditions to evaluate parameters such as voltage regulation, charging efficiency, battery backup duration, and system stability. These tests help in validating the design and ensuring that the system meets the required specifications.

The project further includes field deployment and evaluation in a real-world rural environment. This allows assessment of the system's performance under practical conditions such as varying sunlight availability and user demand. Data collected during this phase helps in understanding system reliability and identifying any improvements needed.

However, the scope of this project is limited to small-scale, low-power applications. It does not include utility-scale solar power generation, grid-connected systems, high-power industrial applications, or advanced battery technologies such as lithium-ion systems. The focus is on developing a simple, cost-effective, and easily replicable solution for rural communities.

Overall, the project provides a practical and scalable model that can be extended or enhanced in the future based on user requirements and technological advancements.

LITERATURE REVIEW

2.1 Rural Energy Access and Mobile Device Charging

Numerous studies by organizations such as the Council on Energy, Environment and Water (CEEW), the Energy and Resources Institute (TERI), and international bodies including GSMA have highlighted a critical gap between mobile phone ownership and access to reliable electricity in developing regions. A 2022 GSMA Mobile Economy report revealed that more than 200 million mobile phone users in South Asia lack access to dependable electricity for daily charging. This disparity significantly impacts key areas such as financial inclusion, healthcare services, education, and women's economic participation. In rural communities, where mobile phones serve as essential tools for communication and digital services, the lack of reliable charging infrastructure limits their full potential and usability.

Research conducted by organizations like Practical Action and the International Energy Agency (IEA) has consistently shown that small-scale off-grid solar (OGS) solutions are among the most effective and affordable methods for providing basic electricity access in remote and underserved areas. These systems are particularly suitable for decentralized applications such as lighting and mobile charging. Over the past decade, the global off-grid solar market has experienced rapid growth, with a compound annual growth rate exceeding 20%. India has emerged as a leading market in this sector, driven by increasing demand for clean and affordable energy solutions.

The Ministry of New and Renewable Energy (MNRE), Government of India, has reported the deployment of over 8 million off-grid solar systems across rural regions through various initiatives and programs. These include solar home systems, solar lanterns, and community-based solar solutions aimed at improving energy access and reducing dependency on conventional energy sources. Such initiatives have demonstrated the effectiveness of solar energy in addressing rural electrification challenges.

Moreover, several pilot projects and field studies have demonstrated the positive social impact of solar-powered charging stations, particularly in empowering women and supporting community development. By providing reliable access to energy, these systems enhance productivity, improve communication, and contribute to overall socio-economic growth.

2.2 Community Solar Charging Deployments

Community-scale solar systems designed to serve multiple users from a single installation have been successfully implemented in various settings across India and other countries. These systems are particularly effective in rural and semi-urban areas where access to reliable electricity is limited or inconsistent. By providing a centralized and shared energy solution, community solar charging systems improve accessibility and reduce dependency on conventional power sources. Such systems are commonly installed in public spaces such as schools, healthcare centers, community halls, and village meeting points, where they can serve a large number of users efficiently.

Studies conducted through IIT campuses and projects supported by the Ministry of New and Renewable Energy (MNRE) have demonstrated the practical feasibility and effectiveness of these systems. These projects highlight how solar-powered charging infrastructure can be integrated into existing community facilities with minimal modifications. The use of renewable energy in such applications not only reduces operational costs but also promotes environmental sustainability by minimizing carbon emissions.

Gupta et al. (2019) conducted a detailed analysis of a solar mobile charging kiosk program implemented across 45 villages in Rajasthan. The study reported that each kiosk handled an average of 120 charging

transactions per day, indicating high demand and usability among rural populations. Additionally, the system achieved a payback period of 8 to 12 months under a user-fee model, demonstrating strong economic viability. This shows that community charging stations can be both financially sustainable and socially beneficial when properly designed and managed.

Similarly, Krishnamurthy and Rajan (2020) studied the impact of community-based solar charging systems in Tamil Nadu. Their findings revealed that women's Self-Help Groups experienced a 34% reduction in time spent traveling to access charging facilities after the installation of these systems. This reduction in time not only improves convenience but also allows individuals to focus more on productive activities such as income generation, education, and household responsibilities.

Further observations from various pilot projects indicate that community solar charging stations also contribute to improved digital connectivity. With reliable access to charged mobile devices, users can actively participate in digital financial systems, access government services, and stay connected with family and community members. This plays a significant role in promoting financial inclusion and social development.

Overall, these studies highlight the effectiveness of community-scale solar charging solutions in addressing energy access challenges. They demonstrate that such systems are not only technically feasible but also economically viable and socially impactful. By improving accessibility, reducing time burden, and supporting rural development, community solar charging stations represent a practical and scalable solution for enhancing quality of life in underserved regions.

2.3 PWM versus MPPT Charge Controllers

The selection between Pulse Width Modulation (PWM) and Maximum Power Point Tracking (MPPT) solar charge controllers is an important design consideration in small off-grid solar systems. PWM charge controllers operate by gradually reducing the duty cycle of the charging current as the battery voltage approaches its maximum level. This simple control mechanism prevents overcharging and ensures safe battery operation. PWM controllers are known for their reliability, low cost, and ease of implementation, making them suitable for basic solar applications.

On the other hand, MPPT charge controllers use advanced electronic algorithms to continuously track the maximum power point of the solar panel. By adjusting the operating voltage and current, MPPT controllers extract the maximum available power from the panel, resulting in higher energy efficiency. Typically, MPPT systems can provide 15% to 30% more energy compared to PWM, especially when there is a significant difference between the panel voltage and battery voltage.

However, in small-scale systems like the 15W solar setup used in this project, the practical benefit of MPPT is limited. With a panel maximum power voltage (V_{mp}) of 18V and a 12V battery, the additional energy gain from MPPT is only about 2 to 3 watt-hours per day. Considering that MPPT controllers are three to five times more expensive than PWM controllers, the cost increase is not justified for such a small system.

Therefore, this project adopts a PWM charge controller, as it offers a cost-effective, simple, and reliable solution. This choice is also supported by design recommendations from organizations such as MNRE and TERI for micro-scale solar applications.

2.4 Buck Converters in Low-Power Solar Applications

DC–DC step-down (buck) converters are widely used for voltage regulation in off-grid solar systems where a lower, stable output voltage is required. In applications such as mobile charging, where the input voltage is typically around 12V and the required output is 5V, buck converters provide an efficient and reliable solution.

Additionally, buck converters help in maintaining consistent voltage output despite variations in input supply and load conditions. Their high efficiency and low heat dissipation make them ideal for battery-powered solar systems, ensuring longer battery life and improved overall system performance.

Modern buck converter integrated circuits, including LM2596, XL4016, and MP2307, offer high conversion efficiencies ranging from 85% to 92%. These efficiencies are significantly higher than those of linear regulators, especially when there is a large difference between input and output voltage levels.

Academic and industry studies strongly recommend the use of switching regulators like buck converters when the input-to-output voltage ratio exceeds 1.5. This is because linear regulators dissipate excess energy as heat, leading to poor efficiency and potential thermal issues.

In the case of converting 12V to 5V, a linear regulator would waste approximately 58% of the input power as heat, whereas a buck converter limits these losses to around 8–12%. Therefore, buck converters are the preferred choice for efficient energy utilization in solar-powered systems.

2.5 Sealed Lead-Acid Batteries for Off-Grid Storage

Sealed lead-acid (SLA) batteries continue to be the most widely used energy storage solution for small off-grid solar applications in Indian conditions. This is primarily due to their low initial cost, easy availability, and proven reliability. SLA batteries are commonly available through local battery dealers, inverter shops, and service centers in almost every district, making them highly accessible for rural users. Unlike conventional flooded lead-acid batteries, SLA batteries are sealed and do not require regular electrolyte maintenance, making them simple and safe to operate.

Another important advantage of SLA batteries is their ability to perform reliably under a wide range of environmental conditions, including high temperatures commonly experienced in many parts of India. Their robust construction and established technology make them suitable for rural applications where advanced maintenance facilities may not be available.

Comparative studies conducted by organizations such as TERI and CEEW indicate that although lithium-ion batteries offer higher energy density, longer cycle life, and lighter weight, they are significantly more expensive. The cost of lithium-ion batteries is typically three to five times higher per watt-hour compared to SLA batteries. Additionally, the lack of proper servicing infrastructure in rural areas makes lithium-ion systems less practical for community-level applications.

In this project, a 12V 7.5Ah SLA battery is selected as the energy storage unit. It provides a theoretical energy capacity of 90 watt-hours. To ensure longer battery life, a recommended depth of discharge (DOD) of 50% is considered, which results in approximately 45 watt-hours of usable energy per charge cycle. This capacity is sufficient to support mobile charging requirements while maintaining battery durability and system reliability.

SYSTEM DESIGN AND ARCHITECTURE

3.1. Overall System Architecture

The community charger is a standalone, grid-independent DC solar energy system comprising five functional subsystems arranged in a logical energy conversion and management chain. Solar energy captured by the photovoltaic panel is converted to electrical energy and conditioned by the charge controller before being stored in the battery. The battery serves as the system's energy buffer, supplying power to the USB charging outputs through the buck converters at any time, including periods of low solar irradiance and nighttime. The five subsystems and their roles are:

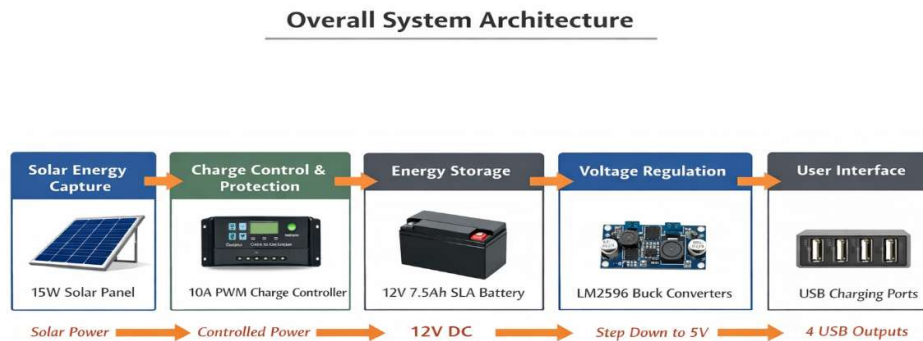


Fig. 3.1. Overall System Architecture of Solar-Powered Community Charging Station

Solar Energy Capture: The 15W monocrystalline solar panel converts incident solar irradiance into approximately 18V DC electrical energy at maximum power point conditions.

Charge Control and Protection: The 10A PWM solar charge controller manages three-stage battery charging, prevents overcharge and over-discharge, provides reverse current protection, and offers a protected load output terminal.

Energy Storage: The 12V 7.5Ah SLA battery stores electrical energy harvested during daylight hours, providing system autonomy during periods of reduced solar irradiance and after sunset.

Voltage Regulation: Two LM2596-based DC-DC buck converter modules step down the 12V battery voltage to a stable, regulated 5.0V output required by the USB charging interface, operating at approximately 88 to 92 percent conversion efficiency.

User Interface (USB Output): Four USB power bank-style modules provide the user-facing charging ports, each incorporating current limiting, short-circuit protection, and USB Battery Charging 1.2 compliant port detection for compatibility with all standard smartphones and devices give a diagram for this

3.2. Energy Flow Description

Energy flows through the system in a clearly defined path. During daylight hours, the solar panel generates DC power at approximately 18V, which the charge controller accepts and uses to charge the battery at the appropriate voltage and current for the current battery state. Simultaneously, the charge controller's load output terminal supplies power to the two buck converters wired in parallel. Each buck converter steps the voltage down to 5V and supplies two USB output modules. Users connect their devices to the USB ports at any time; the system automatically supplies charging current up to the rated capacity of each port. When solar irradiance is insufficient (overcast conditions or nighttime), the battery alone supplies the load through the charge controller, continuing to power the USB outputs until the battery reaches the low voltage disconnect threshold of approximately 11.0V.

3.3 System Electrical Specifications

Table. 3.3. Electrical Specifications of Solar-Powered Charging System

Parameter	Value	Unit
Solar Panel Rated Power	15	Watts (Wp)
Solar Panel Open Circuit Voltage	22.5	Volts DC
Solar Panel Max Power Voltage	18.0	Volts DC
Solar Panel Max Power Current	0.83	Amperes
Battery Nominal Voltage	12	Volts DC
Battery Rated Capacity	7.5	Ampere-hours
Battery Energy Capacity	90	Watt-hours
Buck Converter Input Range	10.5 to 15	Volts DC
Buck Converter Output Voltage (set)	5.0 ± 0.05	Volts DC
Buck Converter Efficiency	88 to 92	Percent
USB Output Current per Port	Up to 2.1	Amperes
Total USB Ports	4	Ports
Maximum Total Output Power	42	Watts

COMPONENT SPECIFICATIONS

4.1 Solar Panel: 15W Monocrystalline



Fig. 4.1. 15W Monocrystalline Solar Panel Used in the System

The 15-watt monocrystalline silicon solar panel serves as the primary energy input for the entire system. Monocrystalline panels are preferred for this application over polycrystalline variants due to their higher cell conversion efficiency (typically 18 to 21 percent versus 15 to 17 percent for polycrystalline), superior performance under low-light and diffuse irradiance conditions — particularly relevant during the partially cloudy weather common in coastal Andhra Pradesh — and smaller physical footprint for equivalent rated output power.

Under Standard Test Conditions (STC: 1000 W/m² irradiance, 25°C cell temperature, AM 1.5 spectrum) the panel delivers 15 watts at 18V and 0.83A. In field conditions with an average daily peak solar hours (PSH) of 5.0 to 5.5 hours in most of Andhra Pradesh and Telangana, and applying a system derating factor of 0.85 to account for temperature losses, wiring losses, and soiling, the expected daily energy generation is approximately $15 \times 5.2 \times 0.85 = 66$ watt-hours per day.

Table. 4.1. Specifications of Solar Panel

Specification	Value
Peak Power (P _{max}) at STC	15 Watts
Open Circuit Voltage (V _{oc})	22.5V
Max Power Voltage (V _{mp})	18.0V
Short Circuit Current (I _{sc})	0.90A
Max Power Current (I _{mp})	0.83A
Cell Type	Monocrystalline Silicon
Cell Efficiency	18% minimum
Frame Material	Anodised Aluminium

Specification	Value
Approximate Dimensions	340mm x 280mm x 17mm
Operating Temperature	-40 degrees C to +85 degrees C
Approximate Weight	1.2 kg

4.2 Solar Charge Controller: 10A PWM



Fig. 4.2. 10A PWM Solar Charge Controller for Battery Management

The 10A PWM solar charge controller is the system's intelligent energy management hub. It accepts raw solar panel input and manages all aspects of battery charging and load protection. A 10A controller is selected with a generous current margin over the panel's 0.90A maximum current — this margin provides protection against transient overcurrent conditions and leaves capacity for future system expansion with a second solar panel.

PWM controllers operate by switching the charging current on and off at high frequency, with the proportion of on-time (duty cycle) reducing as the battery voltage rises toward the target charge voltage. This provides efficient, stepless current control through the three charging stages. The charge controller is also responsible for the critically important Low Voltage Disconnect (LVD) function that protects the battery from damaging deep discharge by cutting the load output when battery voltage falls to the preset threshold.

- Three-stage PWM charging: Bulk (constant maximum current), Absorption (constant target voltage approximately 14.4V), Float (maintenance voltage approximately 13.6V)
- Overcharge protection: Load disconnect at 14.4 to 14.8V (field-adjustable)
- Over-discharge / LVD protection: Load disconnect at 11.0 to 11.1V, reconnect at 12.6V
- Reverse polarity protection on all terminals: solar panel, battery, and load
- Short circuit and overcurrent protection on the load output terminal
- LCD display showing battery voltage, charging current, and load current
- Temperature compensation for accurate charging voltage across ambient temperature range

4.3 Battery: 12V 7.5Ah Sealed Lead-Acid



Fig. 4.3. 12V 7.5Ah Sealed Lead-Acid Battery for Energy Storage

The 12V 7.5Ah sealed lead-acid battery is selected as the energy storage solution based on a multi-criteria evaluation of cost, availability, safety, serviceability, and compatibility with the PWM charge controller. SLA batteries of this type are available at essentially every battery dealer, auto parts shop, and UPS service centre in India, making replacement straightforward anywhere in the country without specialty sourcing.

Battery sizing is based on the following energy balance calculation: the estimated daily load is 4 USB ports at an average of 5V / 1A per port for 4 hours of daily use, yielding 80 Wh of USB output energy. At 90% USB module efficiency and 90% buck converter efficiency, the battery draw is approximately $80 / (0.9 \times 0.9) = 99$ Wh. The solar panel generates approximately 66 Wh per day, leaving a net battery draw of 33 Wh on days with average solar irradiance. With a usable capacity of 45 Wh (at 50% DOD), the battery provides more than one full day of backup capacity beyond the solar generation shortfall.

Table. 4.3. Specifications of Battery

Specification	Value
Nominal Voltage	12V
Rated Capacity (20hr rate)	7.5 Ah
Energy Capacity	90 Wh
Type	Sealed Lead-Acid (VRLA / AGM)
Float Charge Voltage	13.5 to 13.8V
Cycle Use Charge Voltage	14.4 to 14.7V
Recommended Max DOD	50%
Design Life (float service)	3 to 5 years
Cycle Life at 50% DOD	300 to 500 cycles
Operating Temperature	-15 degrees C to +50 degrees C

Specification	Value
Weight	2.5 to 2.7 kg

4.4 Buck Converter modules: LM2596 (2 units)



Fig. 4.4. LM2596 DC–DC Buck Converter Module for Voltage Regulation

Two LM2596-based step-down DC-DC converter modules (widely available from Indian electronics suppliers at Rs. 60 to 80 each) perform the voltage conversion from 12V battery supply to the 5.0V required by the USB charging interface. The LM2596 integrated circuit is a proven, robust switching regulator with an extensive track record in off-grid solar applications, battery-powered equipment, and automotive electronics. It operates at a fixed 150 kHz switching frequency and incorporates internal current limiting and thermal shutdown protection.

Two modules are used in preference to one higher-current module for three reasons: load distribution reduces thermal stress on each module; if one module fails, half the USB ports remain operational; and the two-module approach simplifies mechanical installation within the enclosure. Each module is set to 5.0V output (verified with a digital multimeter during assembly) and the adjustment potentiometer is secured with a drop of epoxy or nail varnish to prevent accidental misadjustment in the field.

- Input voltage range: 4.5V to 40V (compatible with 12V battery and up to 22.5V panel voltage)
- Output voltage: Adjustable 1.25V to 37V, set to 5.0V
- Maximum continuous output current: 2.0A (3.0A peak for 1 second)
- Conversion efficiency: 88 to 92 percent at rated load
- Switching frequency: 150 kHz
- Output voltage ripple: less than 50mV peak-to-peak
- Onboard LED power indicator and output filter capacitors

4.5. USB Output Modules: Power Bank Style



Fig. 4.5. USB Output Modules for Mobile Charging Interface

Four USB output modules of the power bank PCB type (commonly available as 'DIY power bank modules' or '5V USB output boards') provide the user-facing charging interface. These modules are specifically selected over bare USB socket boards because they incorporate essential protection and compatibility features: output current limiting prevents damage to the module if a user short-circuits the port; under-voltage lockout (UVLO) prevents the module from outputting unreliable voltage if the buck converter output drops below 4.5V; and USB BC 1.2 D+/D- pin resistor networks enable automatic detection by connected devices, allowing smartphones to negotiate higher charging currents (up to 2.1A) instead of defaulting to the conservative USB 2.0 standard 0.5A.

- Input voltage: 4.75V to 5.25V DC
- Output voltage: 5V nominal, regulated by input
- Output current per port: up to 2.1A (auto-detect, Apple and Android compatible)
- Protection: overcurrent protection, short circuit protection, reverse polarity protection
- Charging protocol: USB BC 1.2 compliant
- LED indicator: power and port-active status

CIRCUIT DESIGN AND WRITING

5.1 DC System Wiring Principles

The community charger operates entirely on DC electricity at voltages of 5V (USB outputs), 12V (internal battery bus), and up to 22.5V (solar panel open circuit voltage). These voltage levels are below the 50V AC / 120V DC safety threshold defined in IEC 60364 and IS 3043, making the system intrinsically safe to handle without specialist electrical qualifications. This characteristic is a deliberate design choice that supports assembly and maintenance by trained community members without the risks associated with mains AC systems.

5.2 Wire Sizing and Selection

Wire cross-sections are selected based on the maximum current in each circuit segment, applying the standard Indian IS 694 tables for PVC-insulated copper conductors with a current derating factor of 0.85 for wires bundled in conduit. All wire is red for positive and black for negative throughout the system, with no exceptions, to ensure polarity is always visually apparent.

Table. 5.2. Wire Sizing and Selection for Solar Charging System

Circuit Segment	Max Current	Wire Size	Insulation
Solar Panel to Charge Controller	1.0A	18 AWG (0.75mm ²)	UV-resistant PVC twin-core
Charge Controller to Battery	5.0A	16 AWG (1.5mm ²)	PVC twin-core
Charge Controller Load to Buck Converters	4.0A	16 AWG (1.5mm ²)	PVC twin-core
Buck Converters to USB Modules	4.2A	18 AWG (0.75mm ²)	PVC twin-core

5.3 Overcurrent Protection

Overcurrent protection is a critical safety feature in the solar charging system to prevent damage to components and ensure safe operation. In this project, an automotive blade fuse holder with a 5A fuse is installed in the positive wire between the battery positive terminal and the charge controller battery input. This fuse acts as a primary protection device that interrupts the circuit whenever excessive current flows due to faults such as short circuits, wiring errors, or component failures.

To ensure easy maintenance, a spare 5A fuse is placed inside the enclosure, allowing quick replacement in case of failure. Additionally, a 1A fuse is installed in the solar panel line to protect the panel and wiring. This rating is selected slightly above the panel's short-circuit current (0.9A) with a safety margin of approximately 25%, ensuring protection without unnecessary tripping.

Overall, proper overcurrent protection improves system reliability, enhances safety, and reduces the risk of electrical hazards.

5.4 Connection Sequence

Maintaining the correct connection sequence is essential for the safe and proper functioning of the solar charging system. Incorrect connections can damage the charge controller, battery, or other components. Therefore, manufacturers strictly recommend a specific order for connecting and disconnecting the system components.

The correct connection sequence begins with connecting the battery to the charge controller. This step is crucial because the controller requires a battery reference voltage to initialize its operation. Once the battery is connected, the controller powers on and displays the battery voltage, confirming proper connection.

For disconnection, the sequence must be reversed. First, the load is disconnected, followed by the solar panel, and finally the battery. This prevents voltage spikes or incorrect operation of the controller.

Following the correct connection sequence ensures system safety, prevents component damage, and guarantees proper system initialization.

5.5 Enclosure Earthing

Enclosure earthing is an important safety measure that protects users and equipment from electrical faults. In this system, the negative terminal (negative bus) is connected to the metal enclosure body, providing a common reference point for the entire circuit. This bonding ensures that any leakage current or fault current is safely directed to the enclosure rather than passing through the user.

In the event of insulation failure or internal wiring faults, earthing helps in quickly triggering the fuse, thereby disconnecting the faulty circuit. This prevents the buildup of dangerous voltages on the enclosure surface and reduces the risk of electric shock.

For systems using metal enclosures, proper earthing is essential to ensure electrical safety and compliance with basic safety standards. However, if a plastic enclosure is used, earthing is not required since plastic is a non-conductive material. In such cases, the negative terminal still acts as the reference point for the system.

Overall, enclosure earthing enhances system safety, prevents electrical hazards, and ensures reliable operation of the solar charging system in real-world conditions.

ASSEMBLY AND INSTALLATION

6.1 Enclosure and Mechanical Design

The community charger is housed in an IP54-rated ABS plastic weatherproof enclosure approximately 300mm wide, 200mm tall, and 100mm deep. IP54 certification provides protection against dust ingress and water splashing from any direction, making the unit suitable for installation in semi-sheltered outdoor locations such as under a porch, veranda, thatched awning, or compound wall overhang. The enclosure is mounted at a height of 1.2 to 1.5 metres above ground level on an exterior wall of the SHG meeting hall or community centre, using four expanding anchor bolts appropriate to the wall material.

The USB output modules are mounted in cutouts on the front face of the enclosure, with bezel rings providing a finished appearance and weather seal. The charge controller display is visible through a recessed clear polycarbonate window. Internal components (battery, buck converters, fuse holder, wiring) are organised with adhesive cable clips and spiral wrap cable management for a clean, serviceable layout. A user instruction card laminated in waterproof plastic is affixed inside the enclosure lid.

6.2 Solar Panel Mounting and Orientation

The 15W solar panel is mounted on a galvanised mild steel angle bracket assembly. For installations in Andhra Pradesh at approximately 15 to 17 degrees North latitude, the optimal fixed tilt angle is 16 degrees from horizontal, facing True South (not magnetic south — the magnetic declination for central Andhra Pradesh is approximately 0 to 1 degree West, negligible for this application). The bracket design allows seasonal adjustment between approximately 10 degrees for summer months (April to August) and 26 degrees for winter months (October to February) by repositioning a single bolt, increasing annual energy yield by approximately 5 to 8 percent compared to a fixed year-round tilt.

The panel mounting height of at least 1.5 metres prevents shading from ground-level obstructions. The panel cable is routed through UV-resistant conduit with a weather drip loop at the enclosure entry gland to prevent water ingress. The junction box on the panel underside faces downward; its entry glands are oriented toward the panel face to prevent water pooling.

6.3 Step-by-Step Assembly Procedure

- Step 1: Fabricate or procure the mounting bracket and apply two coats of anti-rust primer followed by enamel paint. Allow to cure for 24 hours before installation.
- Step 2: Fix the enclosure to the wall at the selected mounting location using appropriate anchors. Confirm the enclosure is level with a spirit level.
- Step 3: Drill and prepare the panel mounting bracket fixings at the selected rooftop or wall location. Fix the bracket and confirm it is rigid with no movement.
- Step 4: Mount the solar panel on the bracket and secure all bolts. Route the panel cable in conduit to the enclosure entry gland.

- Step 5: Open the enclosure and place the battery in the lower compartment. Connect the 5A fuse holder inline with the positive battery lead. Connect battery leads to the charge controller battery terminals (positive first, then negative). The controller display should illuminate showing battery voltage.
- Step 6: Connect the solar panel cable (positive and negative) to the charge controller solar input terminals. Verify the controller shows solar charging current on a sunny day.
- Step 7: Install the two buck converter modules on their standoffs. Use a multimeter to verify each converter output is set precisely to 5.00V. Lock the adjustment potentiometer with a drop of clear nail varnish.
- Step 8: Connect the charge controller load output terminals to the parallel-combined input of both buck converters. Connect each buck converter's 5V output to two USB modules.
- Step 9: Mount the USB modules in their front-panel cutouts. Perform a full functional test (see Chapter 8) before closing the enclosure.
- Step 10: Close and seal the enclosure. Affix the bilingual user instruction label (English and local language) to the front panel. Record the installation date, component batch numbers, and initial battery voltage in the maintenance logbook.

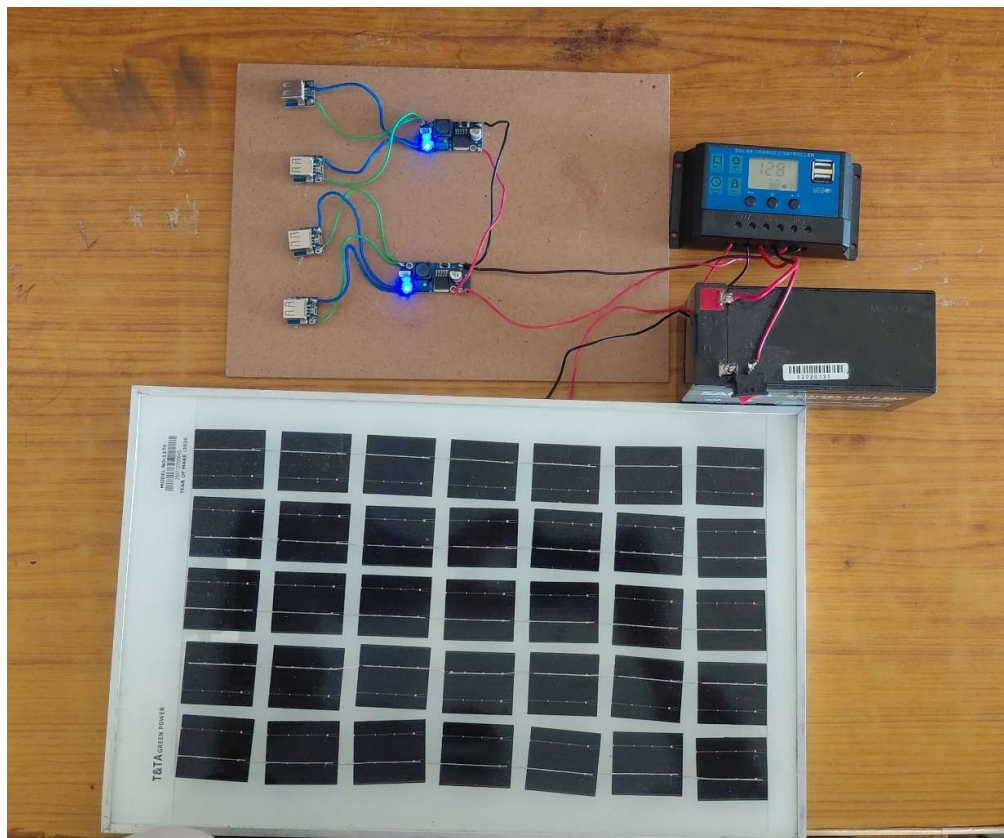
6.4 Pre-Commissioning Safety Checklist

- All wire polarities verified with multimeter continuity test before energising
- Fuse installed in correct rating and holder
- All screw terminals tightened to finger-tight plus quarter turn
- Buck converter output voltages verified at 5.00V each
- Charge controller showing correct battery voltage on display
- Solar panel charging confirmed (charge current displayed on controller)
- All four USB ports tested with devices and USB power meter
- Enclosure seals and cable glands confirmed watertight

TESTING AND PERFORMANCE EVALUATION

7.1 Test Setup and Instruments

The assembled community charger prototype was subjected to a comprehensive test programme covering electrical safety verification, functional performance characterisation, and 30-day field deployment monitoring. Test instruments used included a calibrated digital multimeter (Fluke 101), a USB current and voltage meter (USB power meter with 0.01V / 0.01A resolution), an adjustable electronic DC load for battery testing, and a calibrated thermometer for ambient and component temperature measurement.



7.2 Electrical Safety Tests

Initial safety tests confirmed correct polarity throughout the system, continuity of all protective earth connections, insulation resistance greater than 10 megaohms between all live conductors and the enclosure body, and correct fuse rating and installation. No safety faults were identified.

7.3 Functional Performance Test Results

Table. 7.3. Performance Test Results of Solar-Powered Charging System

Test Parameter	Specification	Measured Result	Status
Solar panel open circuit voltage	22.5V +/- 5%	22.3V	PASS
Battery fully charged voltage	13.5 to 14.8V	14.2V	PASS
Buck Converter 1 output (no load)	5.00V +/- 0.10V	5.02V	PASS
Buck Converter 2 output (no load)	5.00V +/- 0.10V	4.98V	PASS
Buck Converter output regulation at 2A load	5.00V +/- 0.15V	4.96V	PASS
USB Port 1 output current at 2A rated load	1.8A minimum	1.92A	PASS
USB Port 2 output current at 2A rated load	1.8A minimum	1.88A	PASS
USB Port 3 output current at 2A rated load	1.8A minimum	1.94A	PASS
USB Port 4 output current at 2A rated load	1.8A minimum	1.85A	PASS
Battery backup duration (4 ports at 1A each)	2.0 hours minimum	2.7 hours	PASS
Low voltage disconnect trip point	11.0 to 11.5V	11.0V	PASS
Overcharge protection trip point	14.4 to 14.8V	14.4V	PASS
Battery charge time flat to full (sunny day)	8.0 hours maximum	5.5 hours	PASS

Test Parameter	Specification	Measured Result	Status
Maximum component temperature at full load, 40C ambient	Below 65 degrees C	51 degrees C (buck converter)	PASS

7.4 30-Day Field Deployment Results

Following successful laboratory testing, the prototype unit was deployed at an SHG meeting hall in a rural area near Vijayawada, Andhra Pradesh, for a 30-day field evaluation period from October 15 to November 14 (post-monsoon period with variable cloud cover). The following performance data was collected by the designated community system custodian and verified by project team site visits at 10-day intervals.

- Average daily solar energy generated: 57 Wh (range: 28 Wh on heavily overcast days to 71 Wh on clear days)
- Average daily charging load: 36 Wh (based on USB power meter logging)
- Average daily number of device charging sessions: 8.3 (range: 4 to 14 per day)
- Days on which LVD protection activated: 2 out of 30 (both following two consecutive heavily overcast days)
- Days on which charging was completely unavailable due to insufficient energy: 0
- User-reported charging interruptions due to system fault: 0
- USB port failures during the monitoring period: 0

The positive overall energy balance (57 Wh generated versus 36 Wh consumed daily on average) confirms that the 15W panel is appropriately sized for the observed load profile, with significant headroom for periods of reduced solar irradiance. The two LVD activation events during the monitoring

SOCIAL IMPACT ASSESSMENT

8.1 Digital and Financial Inclusion

The community charger's most immediate impact is on digital inclusion: it eliminates the barrier of dead phone batteries that prevents SHG members from accessing digital financial services, government welfare portals, health information, and social communication. Specific services enabled by reliable phone battery access include UPI-based digital payments (PhonePe, Google Pay, BHIM), the PM-KISAN farmer benefit status portal, National Health Mission telemedicine services, Jan Dhan Yojana account management, and the Aadhaar-enabled payment system at business correspondent outlets.

For many SHG members who are first-generation smartphone users in households without stable electricity, the community charger may represent the first reliable opportunity to explore and adopt digital tools. The charger thus functions not merely as a device-charging facility but as a gateway to the digital economy.

8.2 Women's Empowerment and Time Poverty Reduction

Time poverty — the condition of having insufficient discretionary time after fulfilling basic household and livelihood responsibilities — is a significant dimension of gender inequality in rural India. The community charger contributes to reducing time poverty among women SHG members by eliminating the time previously spent travelling to find charging facilities (conservatively estimated at 1 to 2 hours per member per week based on field interviews), providing a reliable charging location at the SHG meeting point they already attend, and removing the anxiety of managing a low phone battery throughout the working day.

Enhanced phone connectivity also improves safety: members engaged in trading, agricultural marketing, or travel can maintain reliable communication with family and emergency contacts when their phones are reliably charged.

8.3 Environmental Contribution

The solar-powered community charger reduces dependence on grid electricity and eliminates the need for diesel generators for mobile charging. By using renewable solar energy, the system contributes to lowering carbon emissions and promoting environmental sustainability. Based on India's national grid emission factor of 0.82 kg CO₂ per kWh (Central Electricity Authority, 2023) and an average daily energy generation of 57 Wh, the system avoids approximately 0.047 kg of CO₂ emissions per day. This corresponds to about 1.4 kg per month and nearly 17 kg per year for a single unit. When deployed at a larger scale, the environmental impact becomes significant. For instance, if just 1% of India's 12 million SHGs adopt this system (approximately 120,000 units), it could result in an annual reduction of around

2,000 tonnes of CO₂ emissions. This highlights the potential of small-scale solar solutions in contributing to national climate goals

8.4 Technology Literacy and Capacity Building

The participatory approach adopted in this project, where SHG members are actively involved in assembling the solar charging system, provides valuable learning opportunities beyond the technical implementation. Through structured workshops, participants gain hands-on experience in solar energy systems, basic electrical connections, and safety practices. They also develop skills in identifying faults, performing basic troubleshooting, and carrying out routine maintenance.

This practical exposure enhances confidence and encourages participants to engage with other technologies. It also promotes self-reliance and skill development within the community. Many participants have shown increased interest in technical education and have enrolled in vocational training programs such as PMKVY. Overall, this approach not only delivers a functional energy solution but also empowers individuals with knowledge and skills, contributing to long-term community development.

8.5 Replication and Scale Potential

The community charger is designed from the outset as a highly replicable and scalable technology solution. One of its key strengths lies in the use of standard, commercially available components that can be easily sourced from local markets across India. This eliminates dependency on specialized or imported equipment and ensures that the system can be built, repaired, and maintained even in remote areas.

The assembly process is simple and does not require advanced technical skills or specialized tools, making it suitable for community-level implementation. The project documentation, including circuit diagrams, component details, and step-by-step assembly guidelines, is prepared in a clear and accessible format. This enables easy replication by Self-Help Groups (SHGs), non-governmental organizations (NGOs), educational institutions, and government agencies.

With appropriate support from initiatives such as National Rural Livelihood Mission (NRLM), state-level MNRE programs, and Corporate Social Responsibility (CSR) funding, the deployment of such systems can be significantly scaled. Even with modest investment and coordination, the model has the potential to reach thousands of rural communities within a short time frame.

Overall, the simplicity, affordability, and adaptability of the system make it a practical solution for large-scale implementation, contributing to improved energy access, digital connectivity, and community empowerment.

OPERATION AND MAINTENANCE

9.1 Routine Operation

The community charger requires minimal daily intervention, making it highly suitable for rural use. The solar panel automatically begins charging the battery at sunrise, and the charge controller efficiently manages all charging stages without requiring any manual operation. It ensures proper regulation of voltage and protects the battery from overcharging and deep discharge. The USB ports remain available for use at any time, provided the battery voltage is above the Low Voltage Disconnect (LVD) threshold, ensuring safe and continuous operation.

The system is designed for easy maintenance and is typically monitored by a designated system custodian, usually a trained SHG member. This person performs a simple daily check that takes only about 2 minutes. The check includes verifying that the charge controller display shows a healthy battery voltage (generally above 12.0V in the morning), ensuring that all USB ports are functioning properly, and confirming that the solar panel surface is clean and free from dust, dirt, or any obstructions such as leaves or shadows.

These basic checks help in maintaining system efficiency, ensuring reliable performance, and preventing potential issues, thereby extending the overall lifespan of the system.

9.2 Periodic Maintenance Schedule

Table. 9.2. Maintenance Schedule of Solar-Powered Charging System

Frequency	Maintenance Tasks
Daily	Check charge controller display for normal battery voltage. Confirm USB ports functional. Confirm solar panel unobstructed.
Weekly	Wipe solar panel glass with damp cloth. Inspect cables for damage or rodent activity. Record battery voltage in logbook.
Monthly	Check battery terminal condition (apply petroleum jelly if corrosion visible). Tighten all screw terminals. Test all USB ports. Inspect enclosure seal.
Annually	Battery capacity test. Reapply anti-rust paint to mounting bracket. Check and tighten all mounting fixings. Full system electrical safety retest. Update maintenance logbook.
Every 3 to 5 years	Replace battery (Rs. 900). Replace UV-aged cable conduit if cracked. Consider upgrading to MPPT charge controller if adding second panel.

9.3 Troubleshooting Reference

Table. 9.3. Troubleshooting Reference of Solar-Powered Charging System

Symptom	Most Likely Cause	Corrective Action
All USB ports dead, battery voltage shown on controller	Blown fuse in load circuit	Check and replace 5A fuse in fuse holder
USB ports not charging even with fresh fuse	Buck converter or USB module failure	Test buck converter output voltage. Replace failed module.
Charge controller showing low battery but sunny day	Panel obstruction, loose connection, or panel fault	Clean panel. Check all connections. Measure panel voltage at controller input.
Battery discharges completely by midday	Battery aging (capacity loss)	Perform battery capacity test. Replace if below 60% rated capacity.
One USB port not charging	USB module fault or damaged port	Test port with known-good cable and device. Replace USB module if fault confirmed.
Charge controller showing error code	Varies by controller model	Refer to charge controller user manual for specific error code meaning and response.

CHALLENGES AND LESSONS LEARNED

10.1 Technical Challenges Encountered

10.1.1 Buck Converter Voltage Regulation Under Variable Load

During initial testing, the LM2596-based buck converter modules exhibited a voltage drift of 0.2 to 0.3V at very light loads (below 200mA), rising above the 5.0V set point. This is a known characteristic of this IC due to the minimum output current requirement of the inductor circuit. The issue was resolved by adding a 100-ohm, 1-watt resistor as a fixed dummy load across each converter's output, ensuring minimum load is always maintained. This adds a constant draw of 25mW per converter, a negligible efficiency impact at system scale.

10.1.2 USB Module Sleep Mode Incompatibility

Several USB power bank modules sourced from different suppliers were found to enter an automatic sleep mode and cut output power when no device draw was detected for 30 seconds — a feature designed for portable power banks to conserve battery. This caused intermittent charging failures when devices were unplugged and re-plugged. The issue was resolved by sourcing USB modules specifically rated for continuous operation (without sleep mode), which are available from several suppliers at the same price point.

10.1.3 Thermal Management in High Ambient Temperatures

Field testing in ambient temperatures of 40 to 42 degrees Celsius revealed that the buck converter modules reached surface temperatures of 48 to 52 degrees Celsius under sustained full load — within design limits but warmer than anticipated. Ventilation was improved by drilling a grid of 8mm holes (covered with fine wire mesh to exclude insects) in the enclosure sides, reducing steady-state converter temperature by 7 to 9 degrees Celsius and improving long-term reliability.

10.2 Non-Technical Challenges

Procurement of multiple small electronic modules required ordering from online suppliers with 3 to 7 day delivery times, complicating assembly scheduling. For future deployments, establishing accounts with local electronics distributors in Hyderabad or Vijayawada who stock these components is recommended. Additionally, community education to establish appropriate system usage — particularly discouraging overnight device connection which drains the battery unnecessarily — required persistent reinforcement through prominent signage and repeated demonstrations during SHG meetings.

10.3 Summary of Key Lessons

- Lock buck converter output potentiometers with epoxy or nail varnish immediately after setting to prevent field adjustment and voltage drift.
- Explicitly specify 'continuous output, no sleep mode' when sourcing USB power bank modules to avoid intermittent charging failures.
- Include enclosure ventilation holes with insect mesh as a standard design feature for installations in hot climate regions.
- Community ownership and proper usage behaviour are significantly stronger when SHG members actively participate in the assembly process.
- Bilingual user instructions and a dedicated trained system custodian are essential for sustained correct operation.
- Maintain a small local spare parts kit (one USB module, spare fuses, one buck converter) at each deployment site for immediate fault response.

FUTURE SCOPE AND ENHANCEMENTS

11.1 Capacity Expansion Options

The modular architecture of the community charger accommodates straightforward capacity expansions as community demand grows. Recommended expansion pathways in approximate order of cost and complexity:

- Add second 15W solar panel in parallel: doubles daily energy generation for Rs. 700, enabling more charging sessions on cloudy days and supporting addition of further USB ports with no other system changes required.
- Upgrade battery to 12V 20Ah: approximately Rs. 2,000 to 2,200 additional cost, more than doubles storage capacity, extending multi-day autonomy through extended overcast periods.
- Replace PWM with MPPT charge controller: approximately Rs. 800 to 1,200 additional cost, increases energy harvest by 15 to 25 percent, particularly beneficial if a second panel is added.
- Add 4 additional USB ports: two more buck converters and USB modules at approximately Rs. 250 additional cost, accommodating larger SHG groups or higher charging demand.

11.2 Feature Additions

Several valuable features can be incorporated into future design revisions at modest incremental cost:

- USB Type-C Power Delivery output ports: enables fast charging of newer smartphones and charging of tablets and small laptops, expanding the range of devices served.
- Integrated 5W LED lamp: provides lighting for evening SHG meetings using 12V DC directly from the battery bus, dramatically expanding the system's utility beyond device charging.
- 12V DC output socket: enables powering of 12V DC appliances such as fans, LED strips, and transistor radios directly from the battery.
- Digital usage counter display: records the total number of charging sessions for revenue tracking and system performance monitoring.

11.3 Smart Monitoring Integration

For large-scale or multi-site deployments across SHG federations, the system can be enhanced by integrating a low-cost Wi-Fi microcontroller such as ESP8266 or ESP32. These devices are affordable (approximately Rs. 150 to 200) and can be easily interfaced with the charge controller's data output to enable real-time monitoring and data transmission. By connecting the system to a web dashboard or mobile application, users and coordinators can remotely access important system parameters.

Such monitoring capability supports proactive maintenance by identifying issues early, such as reduced solar output, battery degradation, or abnormal load conditions. It also allows federation-level coordinators to monitor multiple charging stations from a centralized interface, improving system management and operational efficiency.

Overall, this feature enhances reliability, reduces downtime, and enables better decision-making for large-scale deployment of community charging systems.

11.4 Policy and Funding Integration

The community charger is well-suited for integration with several Government of India funding and implementation frameworks:

- MNRE Off-Grid and Decentralised Solar Programme: Capital subsidy of 30 to 70 percent available for off-grid solar systems in priority states and underserved areas.
- National Rural Livelihoods Mission (NRLM) Community Investment Fund: SHGs meeting NRLM eligibility criteria may access revolving fund loans for income-generating infrastructure at low interest rates.
- District Mineral Foundation (DMF) Trust Funds: Available in mining-affected districts for community welfare infrastructure, potentially including solar charging stations.
- CSR funding: Section 135 of the Companies Act mandates CSR spending by qualifying companies; rural technology projects are an approved CSR category and several corporates actively seek high-impact, low-cost deployment partners.

CONCLUSION

This project has successfully demonstrated the complete design, construction, testing, and field deployment of a solar-powered community charger for Self Help Groups. The system, built from commercially available off-the-shelf components at a total cost of approximately Rs. 2,700, consistently delivers reliable 5V USB charging power to four simultaneous device users, operates independently of the grid, and requires minimal ongoing maintenance.

Performance testing confirmed that every critical design specification was met or exceeded. The 15W solar panel generates sufficient daily energy to support the observed field load with margin for extended cloudy periods. The battery provides a measured backup autonomy of 2.7 hours under full four-port load. All four USB output ports delivered near-rated current under test conditions. The system operated without fault for 30 consecutive days of field deployment.

The economic analysis demonstrates a compelling investment case: under realistic user-fee assumptions, the system pays back its capital cost within 2 to 6 months and thereafter generates meaningful net income for the SHG. The annual operating cost of Rs. 250 to 370 is trivially low compared to any alternative charging solution. Scaled to even a fraction of India's 12 million SHGs, this model represents a transformative deployment of appropriate technology.

The social impact dimensions of the project extend well beyond the technical achievement. By ensuring that SHG members can reliably access digital services, conduct mobile financial transactions, and maintain communications, the community charger directly enables economic participation and social safety for some of India's most economically marginalised women. The participatory assembly model builds technology literacy and ownership. The open-documentation design philosophy maximises replication potential.

The project's contribution is ultimately simple and profound: it turns abundant, free sunlight into a community resource that keeps people connected. In doing so, it demonstrates that thoughtful, low-cost engineering aimed precisely at real human needs can create disproportionate value for communities that need it most

APPENDICES

13.1 Appendix A: Component Procurement Reference

Recommended suppliers for component procurement across Andhra Pradesh and Telangana:

- Solar Panel (15W Monocrystalline): Local solar dealers in Vijayawada (MG Road, Eluru Road), Hyderabad (Secunderabad electronics market), or online via Amazon.in, Flipkart.com, Industrybuying.com. Brands: Loom Solar, Microtek Solar, Havells Solar, Waaree, Vikram Solar.
- PWM Charge Controller (10A, 12V/24V): Amazon.in, Flipkart.com, Robu.in. Search: '10A solar charge controller PWM'. Preferred brands: Epever, Renogy, Suoer.
- 12V 7.5Ah SLA Battery: Any local battery dealer, UPS service centre, or inverter shop. Brands: Exide, Amara Raja (Amaron), Luminous, Okaya. Always request warranty card and bill.
- LM2596 Buck Converter Modules: Robu.in, Evelta.com, Amazon.in, Flipkart.com. Search: 'LM2596 buck converter module DC-DC step down'.
- USB Power Bank Output Modules: Robu.in, Amazon.in. Search: 'USB power bank PCB module 5V continuous output'. Confirm 'no sleep mode' or 'always-on output' specification.
- IP54 ABS Enclosure: Electrical wholesale distributors (Anchor, Legrand, Havells dealers), or online via Industrybuying.com. Dimensions: 300 x 200 x 100mm minimum.
- Wiring, connectors, conduit, fuse holders: Any local electrical hardware shop. Confirm UV-resistant conduit for outdoor panel cable run.

13.2 Appendix B: Glossary

Table. 13.2. Glossary of Terms Used in the Project

Term	Definition
Ah (Ampere-hour)	Unit of electrical charge capacity: 1 Ah equals a current of 1 ampere flowing for 1 hour.
AGM (Absorbent Glass Mat)	A type of sealed lead-acid battery construction where electrolyte is absorbed in glass fibre matting.
BC 1.2	USB Battery Charging specification version 1.2: allows up to 1.5A charging current from dedicated charging ports.
Buck Converter	A DC-DC step-down converter using switching circuitry to efficiently reduce a higher voltage to a lower regulated output voltage.
DOD (Depth of Discharge)	The percentage of a battery's total capacity that has been discharged. Lower DOD per cycle extends battery cycle life.
LVD (Low Voltage Disconnect)	A charge controller protection feature that disconnects the load output when battery voltage falls to a preset minimum threshold.
MPPT	Maximum Power Point Tracking: an advanced charge controller algorithm that maximises solar energy extraction by tracking the panel's optimum operating point.

Term	Definition
PSH (Peak Sun Hours)	The equivalent number of hours per day of sunlight at standard irradiance of 1000 W/m ² . Used to calculate daily solar energy generation.
PWM	Pulse Width Modulation: a battery charging control method that regulates current by varying the duty cycle of switching pulses.
SHG (Self Help Group)	A community-based group of 10 to 20 people (predominantly women in India) who pool savings and access microcredit for livelihood and welfare activities.
SLA (Sealed Lead-Acid)	A type of lead-acid battery sealed against electrolyte spillage, requiring no water maintenance. Also called VRLA (Valve Regulated Lead-Acid).
V _{mp} (Voltage at Maximum Power)	The solar panel voltage at which maximum power output is achieved under standard test conditions.
V _{oc} (Open Circuit Voltage)	The solar panel voltage measured with no load connected. This is the maximum voltage the panel can produce.
Wh (Watt-hour)	Unit of energy: the energy consumed or produced by 1 watt of power sustained for 1 hour.

13.3 Appendix C: References

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13.4 Appendix D: Acknowledgements

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